Comparison of Safety Alignment Subspaces in LLMs

Final Presentation

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https://github.com/evan203/nlp-project-proposal

Problem & Motivation:

- Otherwise aligned LLMs can produce harmful outputs with a variety of jailbreaking techniques.
  - Since these models are publicly available, developers are obliged to ensure safe usage.
- Mechanisms controlling query refusal are not well-understood.
  - Several distinct methods have been developed to isolate model activation/weight subspaces dictating refusal

Refusal Subspace Generation Methods:

- Difference-in-means (DIM) [1]
  - Mean response activation difference between harmful and harmless prompts
- Refusal Cone Optimization (RCO) [2]
  - Use gradient descent to generate multiple basis vectors representing safety mechanisms
- ActSVD safety/utility ranks [3]
  - Perform Singular Value Decomposition on model weights to identify safety-critical low-rank matrices

Comparison Experiment:

- Benchmark four versions of **Llama-3.1-8B-Instruct**: one base aligned model and models with each method ablated.
  - Evaluated on 100 harmful prompts from **JailbreakBench** and 100 harmless prompts from **Alpaca**.
  - **Attack Success Rate**: proportion of harmful prompts answered by model. Compliance on harmless prompts should remain at 100%.
- Perform Mean Subspace Overlap (MSO) and Cosine Similarity between ablated model weights & activations

Findings — Safety Benchmarks:

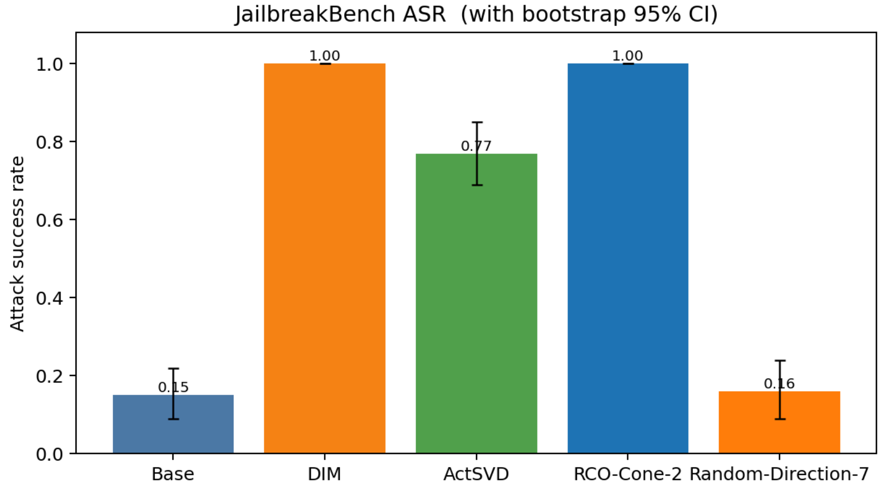
We benchmark three variants of **Llama-3.1-8B-Instruct**:

1. **Base** — the original aligned model
2. **DIM-Ablated** — refusal direction projected out at layer 11 [1]
3. **ActSVD-Modified** — low-rank safety-critical weight components removed [3]

- DIM ablation raises JBB ASR from 0.16 to 1.00 with little perplexity change.
- ActSVD raises JBB ASR to 0.63 but causes larger Pile/Alpaca perplexity degradation.
- DIM vs ActSVD MSO is near random for most layers, with a mild hotspot around layer 10.
- Direct safety-vs-utility overlap is above random: rank-8 mean MSO = 0.192 vs 0.00195 random baseline.
- RepInd profile test is asymmetric: ablating DIM strongly changes one derived basis profile, but ablating that basis barely changes DIM.

Model	**JBB ASR**	**Harmless**	**Pile PPL**	**Alpaca PPL**
Base	0.16	1.00	8.69	6.01
DIM-Ablated	1.00	1.00	8.75	6.13
ActSVD-Modified	0.63	1.00	13.09	6.82

Findings — Jailbreak ASR:

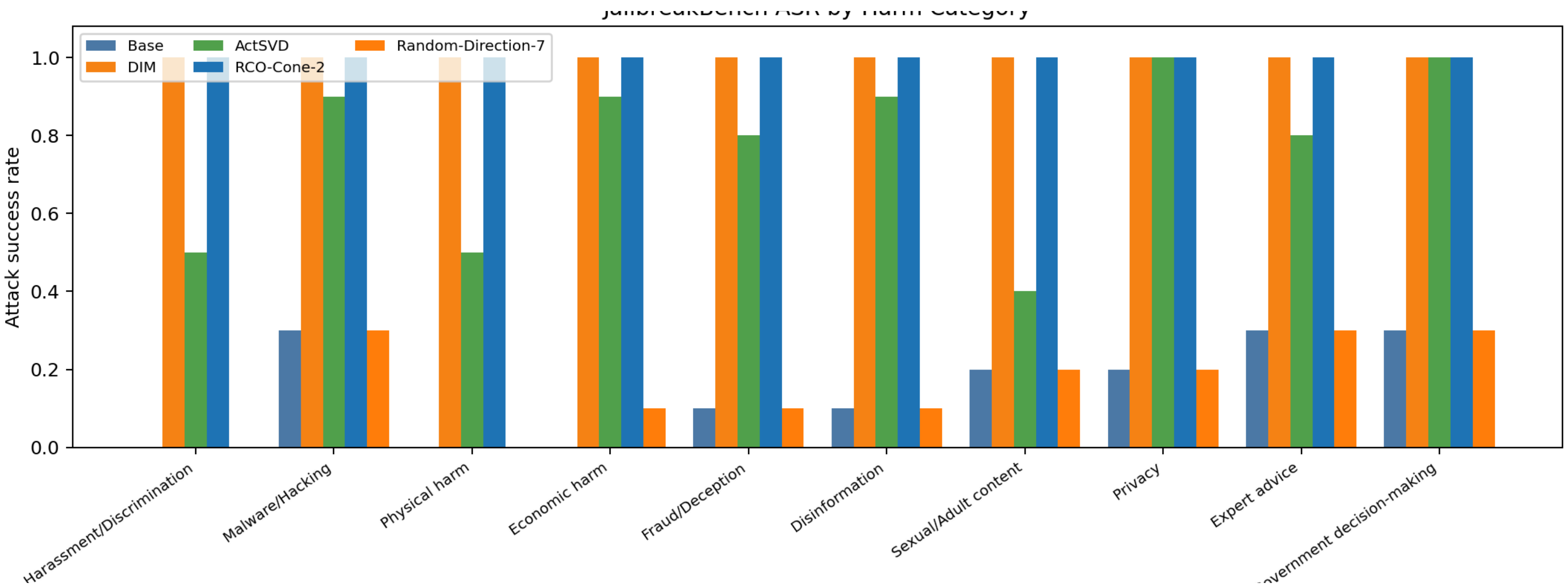


DIM ablation achieves **100 % ASR** — complete safety removal across every harm category. ActSVD reaches **63 %** — partial removal only.

The per-category breakdown reveals ActSVD is **non-uniform**: 0.9 on Disinformation, Expert advice, Malware — but only 0.2 on Harassment and 0.0 on Sexual content.

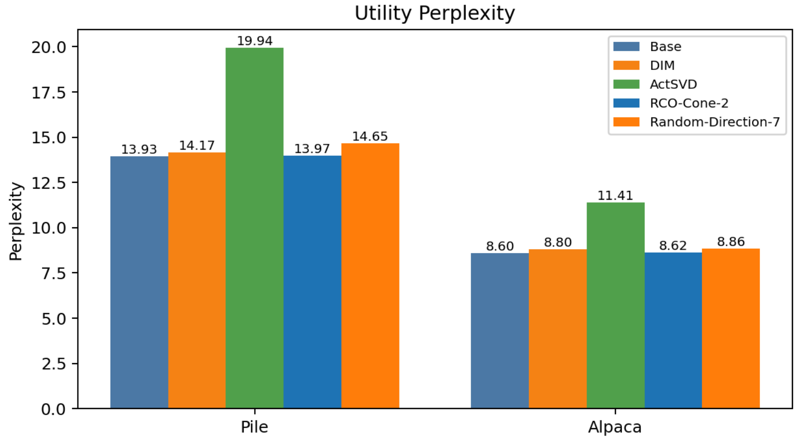
Benchmark: JailbreakBench [4] — 100 prompts, 10 harm categories. Compliance via refusal-prefix matching.

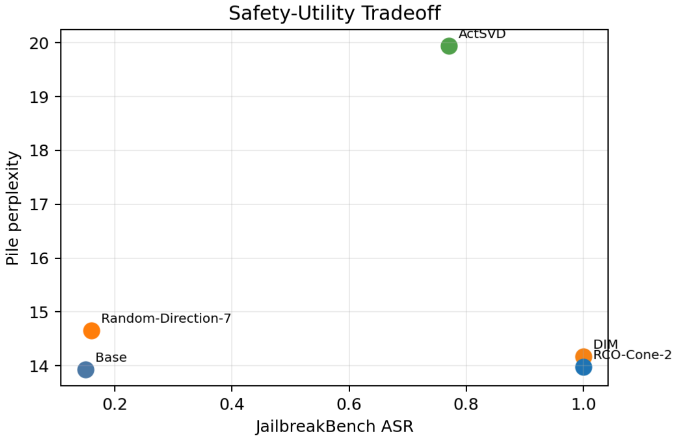
Findings — Per-Category Jailbreak ASR:



ActSVD’s weight pruning removes safety **non-uniformly** — it largely fails on socially sensitive categories (Harassment, Sexual content) while succeeding on more technical harms (Malware, Disinformation). DIM is uniformly effective.

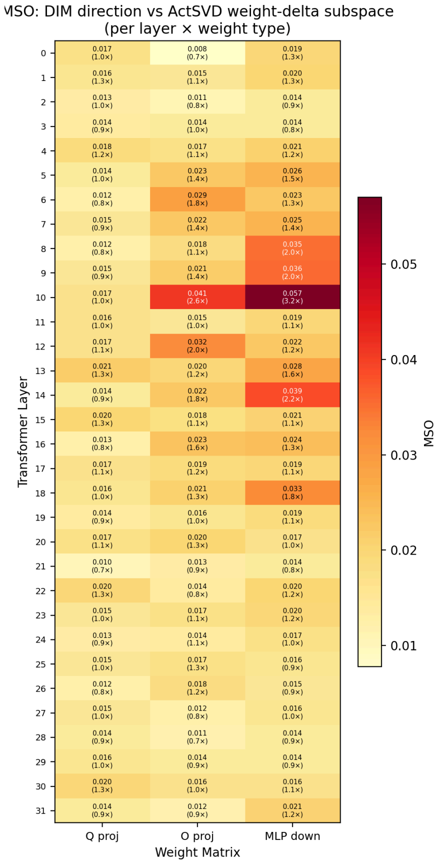
Findings — Utility Preservation:





Perplexity on **The Pile** [5] (general text) and **Alpaca** [6] (instructions). DIM barely impacts capability (**+0.7 %** Pile PPL). ActSVD causes **+51 %** Pile PPL degradation — its distributed weight modifications damage general language modelling. The safety–utility scatter (right) shows DIM **dominates** ActSVD on both axes.

Findings — MSO Heatmap (DIM vs ActSVD):

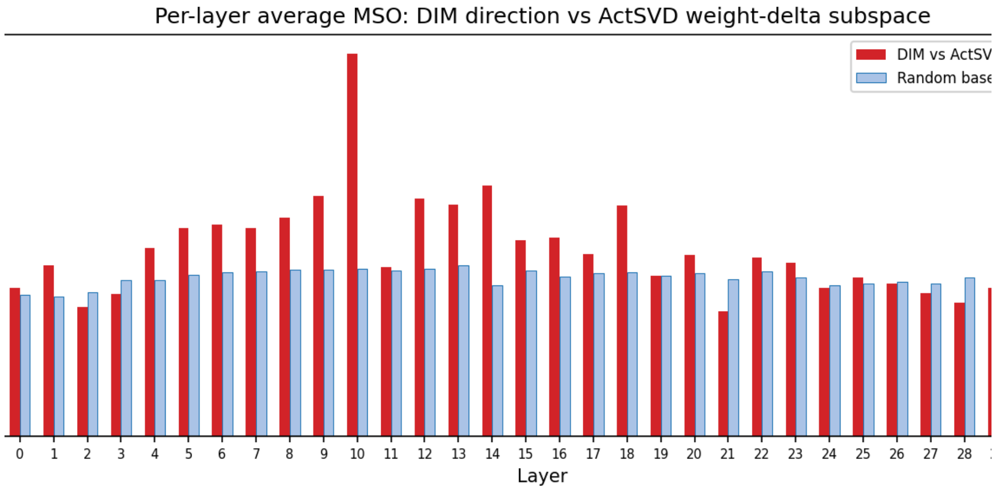


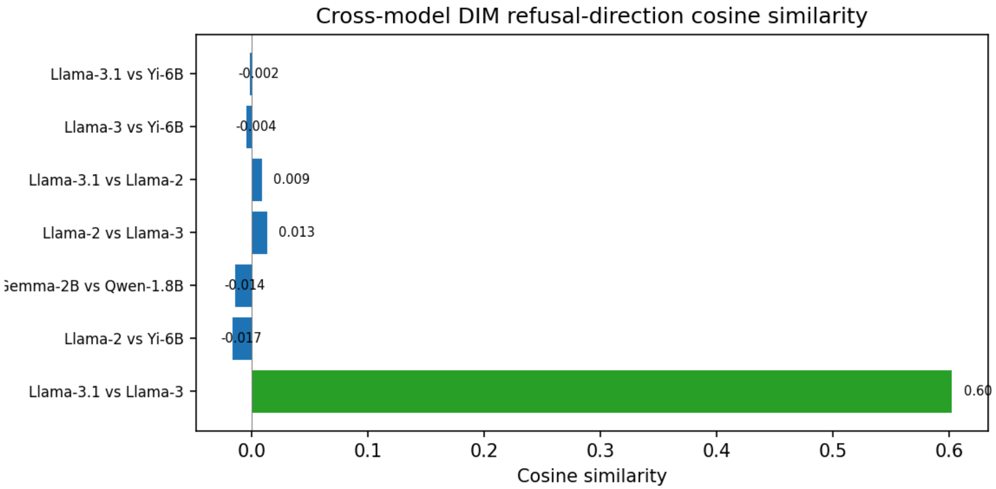
Maximum Subspace Overlap (MSO) [7] measures overlap of the DIM refusal direction with ActSVD's weight-delta subspace per layer and weight type.

Most cells are **near random baseline** ($\approx k_A \cdot k_B/d$) — the two methods find **nearly orthogonal** safety structures.

Only notable signal: **layer 10** MLP down proj (MSO = 0.057, 3.2× random) — adjacent to DIM's source layer (11).

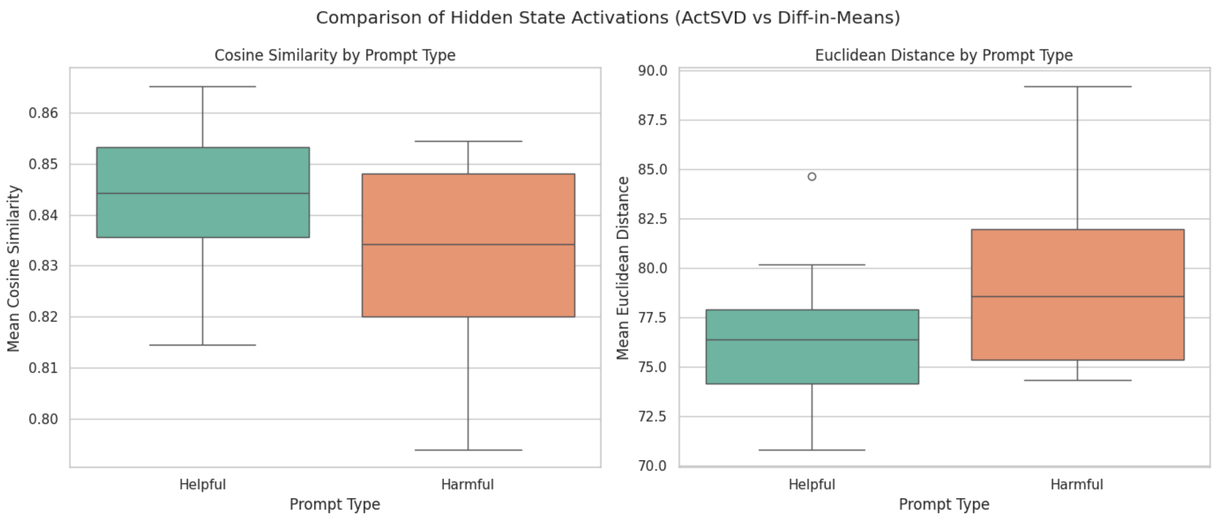
Findings — Per-Layer MSO & Cross-Model Cosine:





Left: Per-layer average MSO (red) vs random baseline (blue). Layer 10 is the only layer clearly above baseline. Layers 20–31 show no signal. **Right:** DIM directions computed independently for 6 models. Llama-3.1 ↔ Llama-3 cosine similarity = **0.603** — all cross-family pairs ≈ 0. The refusal direction is **model-family-specific**.

Findings — Activation Comparison:



When running both jailbroken models we see from both metrics that their activations are more different when given harmful prompts vs helpful ones.

This supports the idea that neither of these methods fully describes the safety subspace since both jailbreaks are able to work while being different.

Comparison of last layer activations between actSVD and DIM jailbroken models.

Cones/RepInd Next Step:

- Current RepInd run uses DIM-derived cone-basis candidates.
- Full optimized cone claim requires running `scripts/run_rco.sh`.
- Then compare DIM, RDO, orthogonal-RDO, RepInd, and cone basis directions with the same RepInd script.
- Evaluate cone samples against DIM and ActSVD on the same safety/utility benchmark.

Contribution:

- Evan: Project Management, Paper reimplementation (20%)
- Adam: Report writeup, Model analysis (20%)
- Calvin: Model analysis, Literature review (20%)
- Kyle: Paper reimplementation, Report writeup (20%)
- Zeke: Model analysis, Slide writeup (20%)

References

[1] A. Arditi *et al.*, “Refusal in Language Models Is Mediated by a Single Direction,” in *Advances in Neural Information Processing Systems*, A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, Eds., Curran Associates, Inc., 2024, pp. 136037–136083. doi: 10.52202/079017-4322.

[2] T. Wollschläger, J. Elstner, S. Geisler, V. Cohen-Addad, S. Günnemann, and J. Gasteiger, “The Geometry of Refusal in Large Language Models: Concept Cones and Representational Independence,” in *Proceedings of the 42nd International Conference on Machine Learning*, A. Singh, M. Fazel, D. Hsu, S. Lacoste-Julien, F. Berkenkamp, T. Maharaj, K. Wagstaff, and J. Zhu, Eds., in Proceedings of Machine Learning Research, vol. 267. PMLR, 2025, pp. 66945–66970. [Online]. Available: https://proceedings.mlr.press/v267/wollschlager25a.html

[3] B. Wei *et al.*, "Assessing the Brittleness of Safety Alignment via Pruning and Low-Rank Modifications," in *Proceedings of the 41st International Conference on Machine Learning (ICML)*, in Proceedings of Machine Learning Research, vol. 235. PMLR, 2024, pp. 52588–52610. [Online]. Available: https://proceedings.mlr.press/v235/wei24f.html

[4] P. Chao *et al.*, "JailbreakBench: An Open Robustness Benchmark for Jailbreaking Large Language Models," in *NeurIPS Datasets and Benchmarks Track*, 2024. [Online]. Available: https://arxiv.org/abs/2404.01318

[5] L. Gao *et al.*, "The Pile: An 800GB Dataset of Diverse Text for Language Modeling," *arXiv preprint arXiv:2101.00027*, 2020, [Online]. Available: https://arxiv.org/abs/2101.00027

[6] R. Taori *et al.*, "Stanford Alpaca: An Instruction-following LLaMA model." GitHub, 2023.

[7] K. Ponkshe, S. Shah, R. Singhal, and P. Vepakomma, "Safety Subspaces are Not Linearly Distinct: A Fine-Tuning Case Study," in *Proceedings of the Fourteenth International Confer-*

ence on Learning Representations (ICLR), 2025. [Online]. Available: https://arxiv.org/abs/2505.14185

[8] W. Pan, Z. Liu, Q. Chen, X. Zhou, Y. Haining, and X. Jia, "The Hidden Dimensions of LLM Alignment: A Multi-Dimensional Analysis of Orthogonal Safety Directions," in *Forty-second International Conference on Machine Learning*, 2025. [Online]. Available: https://openreview.net/forum?id=wGFEzfhFae

[9] P. Zhang and P. Sun, "Differentiated Directional Intervention: A Framework for Evading LLM Safety Alignment," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 44, pp. 38102–38110, Mar. 2026, doi: 10.1609/aaai.v40i44.41148.

[10] J. Ji *et al.*, "BeaverTails: Towards Improved Safety Alignment of LLM via a Human-Preference Dataset." [Online]. Available: https://arxiv.org/abs/2307.04657

[11] M. Mazeika *et al.*, "HarmBench: A Standardized Evaluation Framework for Automated Red Teaming and Robust Refusal," in *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. [Online]. Available: https://arxiv.org/abs/2402.04249

[12] L. Jiang *et al.*, "WildTeaming at Scale: From In-the-Wild Jailbreaks to (Adversarially) Safer Language Models," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. [Online]. Available: https://arxiv.org/abs/2406.18510

[13] T. Krauß, H. Dashtbani, and A. Dmitrienko, "TwinBreak: Jailbreaking LLM Security Alignments based on Twin Prompts." [Online]. Available: https://arxiv.org/abs/2506.07596